Buildroot Additional Hints

Advanced Embedded Software Development

with **Dan Walkes**



University of Colorado **Boulder**

Learning objectives:

Understand tips for optimizing Buildroot development workflow

Buildroot Dependencies

```
Symbol: BR2_PACKAGE_BASH [=n]
Type  : bool
Prompt: bash
  Location:
    -> Target packages
(1)   -> Shell and utilities
  Defined at package/bash/Config.in:1
  Depends on: BR2_USE_MMU [=y] && BR2_PACKAGE_BUSYBOX_SHOW_OTHERS [=n]
  Selects: BR2_PACKAGE_NCURSES [=n] && BR2_PACKAGE_READLINE [=n]
  Selected by [n]:
  - BR2_SYSTEM_BIN_SH_BASH [=n] && <choice> && BR2_USE_MMU [=y] && BR2_PACKAGE_BUSYBOX_SHOW_OTHERS [=n]
  - BR2_PACKAGE_LINUX_TOOLS_SELFTESTS [=n] && BR2_LINUX_KERNEL [=y] && BR2_PACKAGE_BUSYBOX_SHOW_OTHERS [=n] && BR2_USE_MMU [=y]
  - BR2_PACKAGE_V4L2LOOPBACK_UTILS [=n] && BR2_PACKAGE_V4L2LOOPBACK [=n] && BR2_USE_MMU [=y]
  - BR2_PACKAGE_DEHYDRATED [=n] && BR2_USE_MMU [=y]
```

"Depends on" items are missing

```
Shell and utilities
Arrow keys navigate the menu.  <Enter> selects submenus ---> (or
submenus ----).  Highlighted letters are hotkeys.  Pressing <Y>
selects a feature, while <N> excludes a feature.  Press <Esc><Esc>
exit, <?> for Help, </> for Search.  Legend: [*] feature is selec

        *** Shells ***
    [ ] mksh
    [ ] zsh
        *** Utilities ***
    [ ] at
    [ ] ccrypt
    [ ] dialog
    [ ] dtach
    [ ] easy-rsa
    [ ] file
    -(+)

    <Select>   < Exit >   < Help >   < Save >   < Load >
```

Why is bash missing from the shells list?

```
Symbol: BR2_PACKAGE_BUSYBOX_SHOW_OTHERS [=n]
Type  : bool
Prompt: Show packages that are also provided by busybox
  Location:
    -> Target packages
(1)   -> BusyBox (BR2_PACKAGE_BUSYBOX [=y])
  Defined at package/busybox/Config.in:33
  Depends on: BR2_PACKAGE_BUSYBOX [=y]
  Selected by [n]:
```

Buildroot Cross Compiling

- “Brute Force” way to compile new source code:
  - push to git
  - Change FOO_VERSION hash

```
AESD_ASSIGNMENTS_VERSION = 6b8091a89a9b2d77a8da00804fdf8dc2f012d214
AESD_ASSIGNMENTS_SITE = git@github.com:cu-ecen-5013/assignments-complete-private.git
AESD_ASSIGNMENTS_SITE_METHOD = git
```

- This is slow, results in a lot of commits when debugging.
- Is there a better way?

Buildroot Cross Compiling

- Use FOO_OVERRIDE_SRCDIR
  - From the buildroot directory, run:
  - *make AESD_ASSIGNMENTS_OVERRIDE_SRCDIR=~/aesd-assignments/ aesd-assignments-rebuild all*
  - Ignores the version specified in your aesd-assignments.mk file
  - Uses the source as it exists in the specified source directory
  - Use all caps version of your module name before _OVERRIDE_SRCDIR
- Use <packagename>-dirclean to start from an empty build output dir

https://buildroot.org/downloads/manual/manual.html#_advanced_usage
https://buildroot.org/downloads/manual/manual.html#pkg-build-steps

Buildroot Cross Compiling

```
dwalkes@AESD-001:~/buildroot-complete-private-rpi/buildroot$ make AESD_ASSIGNMENTS_OVERRIDE_SRCDIR=~/assignments-complete-private aesd-assig
nments-rebuild all
rm -f /home/dwalkes/buildroot-complete-private-rpi/buildroot/output/build/aesd-assignments-custom/.stamp_rsynced
rm -f /home/dwalkes/buildroot-complete-private-rpi/buildroot/output/build/aesd-assignments-custom/.stamp_staging_installed
rm -f /home/dwalkes/buildroot-complete-private-rpi/buildroot/output/build/aesd-assignments-custom/.stamp_target_installed
rm -f /home/dwalkes/buildroot-complete-private-rpi/buildroot/output/build/aesd-assignments-custom/.stamp_images_installed
rm -f /home/dwalkes/buildroot-complete-private-rpi/buildroot/output/build/aesd-assignments-custom/.stamp_host_installed
rm -f /home/dwalkes/buildroot-complete-private-rpi/buildroot/output/build/aesd-assignments-custom/.stamp_built
>>> aesd-assignments custom Syncing from source dir /home/dwalkes/assignments-complete-private
rsync -au --chmod=u=rwX,go=rX  --exclude .svn --exclude .git --exclude .hg --exclude .bzr --exclude CVS /home/dwalkes/assignments-complete-p
rivate/ /home/dwalkes/buildroot-complete-private-rpi/buildroot/output/build/aesd-assignments-custom
```

- Uses rsync to synchronize content between your OVERRIDE_SRCDIR and buildroot/output directory

https://buildroot.org/downloads/manual/manual.html#_advanced_usage